INDIVIDUAL PROJECT REPORT

AI in Healthcare

A Connected AI Approach for Early Tuberculosis Detection in India

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Group	Group 03
Project Focus	AI-supported TB detection and coordinated care

Prepared from the Group 03 presentation. Sunita is a fictional case; existing building blocks are distinguished from the group's recommendations.

Course / Institution: _____

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1. Executive Summary and Problem Statement

This report examines how artificial intelligence (AI) could help India's healthcare system detect tuberculosis (TB) earlier, support safer clinical decisions and use limited resources more effectively. It is based on the Group 03 presentation and its fictional patient, Sunita, a 42-year-old farm worker with a three-week cough, weight loss and evening fevers. Her nearest health centre has no radiologist, records may be on paper or separated across systems, and every referral may require another journey. This case is illustrative, not a real patient record.

The central issue is a connected-care gap. AI-based chest X-ray reading, digital health records and referral services may exist as separate building blocks, but the presentation found no public evidence that they operate as one complete loop for triage, referral, follow-up and resource planning. This report proposes a connected system rather than claiming that the full system is already deployed.

The proposed loop combines patient and provider data; secure quality checks; AI models and programme rules; risk flags and referral suggestions; clinician-led decisions; and monitoring of accuracy, fairness, drift and outcomes. A high TB-likelihood score should prompt consideration of a confirmatory test, not be treated as a diagnosis.

Objectives

• Describe the problem and context using figures stated in the presentation.
• Explain the proposed integrated workflow and relevant AI techniques.
• Apply the workflow to Sunita's illustrative patient journey.
• Set out a phased pilot and measurable evaluation plan.
• Identify ethical, privacy, regulatory and operational safeguards.

Scope and evidence boundary

The report uses the supplied presentation as its primary project source. It distinguishes existing examples—such as AI X-ray reading in India's TB programme—from new proposals such as unified rules, a knowledge graph, follow-up tracking and cross-facility resource forecasting. AI cannot eliminate diagnostic error or guarantee outcomes.

2. Why Tuberculosis Is a Suitable Starting Use Case

The presentation selects TB because the burden is high and delayed detection can affect both individual patients and the wider community. It reports that India accounted for about 25% of the world's TB cases in 2024 and 32% of drug-resistant TB cases. It also notes a positive trend: TB incidence had fallen by 21% since 2015. Progress does not remove the remaining access gap.

The presentation estimates that around one lakh TB cases in 2024 were not reaching the national programme. These are people who may not be diagnosed or linked to treatment in time. Sunita's fictional case illustrates several barriers at once: symptoms need assessment, X-ray interpretation may be delayed, records may not follow the patient, and referrals can require travel, time and money.

Access to specialists is another constraint. The presentation cites Rural Health Statistics data up to March 2022 showing a 79.5% shortfall of specialists at rural Community Health Centres (CHCs), and that 541 of 5,480 CHCs had all four basic specialists. These figures are older than the current date and should be understood as evidence of a structural challenge, not a real-time count.

India also has a growing digital health foundation. The presentation states that, as of July 2026, more than 104 crore health records were linked to more than 93 crore ABHA accounts. It also notes that AI-enabled handheld X-rays have been used to find TB cases and that WHO recommended computer-aided X-ray reading for TB screening in 2021. Digital records alone do not guarantee interoperability, consent or follow-up.

Project opportunity

Connect the building blocks into a reliable pathway: link an X-ray risk flag to an approved confirmatory-testing process, track the referral, return results to the appropriate care team with consent, and use aggregate demand forecasts to organise X-ray, laboratory and test-kit capacity. This is the group's proposal, not a claim that the complete system is already deployed across India.

Why start small?

A few districts with digital X-ray capability and basic connectivity are a more practical starting point than a nationwide launch. The pilot can test data quality, clinician workload, referral completion and resource planning before expansion. Facilities without reliable digital records or connectivity may need infrastructure improvements first.

3. Proposed Integrated AI Approach

The proposed system is a connected loop from patient information to clinician-led action and then to monitoring. Its purpose is to support early detection, coordinated care and resource planning while keeping people accountable for decisions.

Step 1 — Patient and provider data. Relevant inputs may include chest X-rays, symptoms, laboratory results and existing health records. ABHA-linked records may support continuity where linkage is available and consent and applicable requirements are respected. Only data needed for the use case should be collected.

Step 2 — Secure quality checks. Check missing or inconsistent fields, timestamps, duplicate records and identity matching. Access should be role-based and important actions logged. If essential information is missing or a connection fails, mark the output unavailable rather than interpreting it as low risk.

Step 3 — Models and knowledge. Image AI analyses a chest X-ray and produces a TB-likelihood score. Supervised machine learning may support risk estimation or demand forecasting. Approved programme rules can define when a high score should prompt confirmatory testing. A knowledge graph can connect symptoms, tests, medicines and facilities using shared vocabulary. Forecasting may estimate X-ray, laboratory and test-kit demand.

Step 4 — Alerts and suggestions. Present the risk flag, uncertainty, relevant data and suggested next step under approved protocols. Do not present a score as a definitive diagnosis or prescribe treatment automatically. A referral suggestion should identify the responsible team and allow review of supporting information.

Step 5 — Human decision. A doctor or authorised health professional reviews the X-ray, symptoms, history and AI output. The clinician can order a confirmatory test, override a suggestion or request more information. AI must not delay urgent care or replace clinical judgement.

Step 6 — Monitor and improve. Record performance, fairness, alert burden, referral completion and outcomes. Changes to a model or rule should happen through controlled review, validation and approval—not automatic self-modification. The full loop is the group’s proposal.

Examples of existing building blocks

The presentation names Apollo’s AICVD cardiac risk model, Siemens Healthineers AI-Rad Companion and Practo’s ABDM-compliant records software as examples. They are examples only, not endorsements and not the group’s proposed integrated product.

4. AI Techniques and Their Roles

Different techniques serve different tasks. Selection should depend on purpose, available data, validation and the consequences of error—not on how advanced the technology sounds.

Technique	Role	Limitation
Deep learning	Reads chest X-ray images and outputs a TB-likelihood score.	Performance varies by setting and threshold; local validation is required.
Supervised ML	Learns from labelled cases to estimate risk and forecast demand.	Needs representative data, including rural patients.
Probabilistic reasoning	Shows likelihood with uncertainty, not only yes/no.	Scores need calibration and can be over-trusted.
Expert rules	Encodes programme rules, e.g. high score prompts a confirmatory test.	Transparent but rigid; rules need updating.
Knowledge graph + ontology	Links symptoms, tests, drugs and facilities through shared meanings.	Costly to build and maintain.
Generative AI	Drafts referral summaries from approved records.	May hallucinate or omit facts; a doctor must verify.

The presentation distinguishes prediction, decision support, optimisation and text generation. Prediction produces a score; decision support suggests a next step; optimisation can plan resources or routes; and generative AI drafts text. A* search could be considered for routing a mobile X-ray van, but that is an optimisation task, not a diagnostic method.

The speaker notes cite a WHO triage benchmark of at least 90% sensitivity and 70% specificity and mention a review reporting sensitivity of 0.88–0.91 for some AI tools. These are contextual benchmarks from the presentation, not guaranteed performance for a new model or pilot. Local performance must be measured before deployment.

A generative-AI study mentioned in the presentation reported 1.47% hallucinated statements and 3.45% omitted statements. These figures reinforce the need for source-linked summaries and human verification; they are not universal error rates for all systems.

5. Sunita’s Illustrative Patient Journey

Sunita is a fictional 42-year-old farm worker with a three-week cough, weight loss and evening fevers. This journey shows how the proposed workflow could help coordinate care. It is an example, not a diagnosis or real patient record.

Stage	What happens	Status
1. Village visit	A health worker records symptoms and takes a digital chest X-ray. The record is linked to her ABHA ID with consent.	Digital X-ray and records exist in some settings.
2. AI flag	Image AI provides a TB-likelihood score with uncertainty. A high score prompts consideration of a confirmatory test.	Connected AI flag is proposed.
3. Doctor review	A district doctor reviews the X-ray, score and history, can override the output and decides whether to order a molecular test.	Proposed connected workflow.
4. Follow-up	The referral is booked, reminders are sent and results are shared with the local clinic with consent.	Follow-up tracking is proposed.

The AI does not diagnose TB on its own. A high likelihood score is a triage signal, and confirmatory testing and clinical assessment remain important. A clinician must consider the full presentation and other possible causes. Urgent assessment must not be delayed while waiting for an AI output.

Parallel district-planning use case

District planners could use forecasts of X-ray, laboratory and test-kit demand, together with information about follow-up gaps, to help plan where mobile vans, staff and supplies are needed. Forecasting methods should be selected by testing local historical data. Planners should compare forecasts with actual demand and account for uncertainty, seasonal patterns and changes in service access.

Expected benefit and boundary

If locally validated and properly integrated, the proposed workflow may reduce delays in interpreting X-rays, make referrals easier to track and support resource allocation. These are expected benefits to test, not guaranteed outcomes. The system cannot eliminate diagnostic error, remove every travel barrier or guarantee treatment completion.

6. Implementation Roadmap and Evaluation

The group proposes a phased implementation. Each phase should have a decision gate so that the system advances only when evidence supports safety, fairness, workflow fit and cost-effectiveness.

Phase	Activities	Decision gate
1. Define	Select use case and district partners. Map current pathway and measure baseline waiting times, referrals and resource use.	Baseline definitions and owners agreed.
2. Build	Choose model and rules; test retrospectively. Check data quality, sensitivity, specificity and subgroup performance.	Meets pre-agreed validation criteria.
3. Pilot	Start in a few suitable facilities. Doctors remain in charge; record outputs, overrides, referrals and failures.	Safe and reliable workflow.
4. Evaluate	Review safety, fairness, workflow, cost, staff feedback and operational measures.	Evidence supports benefit without unacceptable harm.
5. Expand	Extend only to sites with digital X-ray, connectivity, training and governance. Continue monitoring.	Ongoing accountable ownership.

What should be measured?

1. Detection: sensitivity, specificity, false alarms and missed cases. The presentation cites a WHO triage benchmark of at least 90% sensitivity and 70% specificity; the pilot must measure actual local performance.
2. Timeliness: days from first visit to confirmatory test.
3. Follow-up: share of patients completing referrals and follow-up visits.
4. Resource use: X-ray and laboratory waiting times and forecast error compared with the current method.

The presentation states that no reliable public baseline was found for measures 2–4. Phase 1 must measure current performance. Compare results with the same site’s “before” period and define success criteria before examining outcomes. Also record workload, alert burden, privacy incidents and differences in performance across region, sex and age.

Evaluation must include negative findings. If the model misses cases, creates excessive false alarms, performs poorly for rural patients or adds unsustainable workload, revise or stop the project rather than expanding automatically.

7. Risks, Ethics and Governance

The proposed system uses sensitive health information and may influence which patients receive further assessment. Safeguards must be built into the design and maintained throughout the pilot.

Risk	Safeguard
Biased or limited data	Validate by region, sex and age; include rural and underserved patients where feasible. Report limitations where samples are small.
False alarms or missed cases	Track false positives and false negatives; use confirmatory testing and clinician review. Never treat the AI score as a diagnosis.
Over-reliance on AI	Show uncertainty and reasons; allow clinicians to override outputs and log overrides.
Privacy and consent	Use purpose-limited access, consent where required, role-based permissions and audit logs. Review applicable ABDM and DPDP requirements.
Medical AI regulation	Review applicable CDSCO medical-device-software guidance and ICMR’s 2023 AI ethics guidelines before implementation.
Model drift	Monitor performance over time; investigate changes and retrain/retest only through formal approval.
Cost and infrastructure	Plan for devices, connectivity, training, maintenance and regulatory approval. Start where digital X-ray capability exists.

Human accountability

A named clinical lead should own the use case, while technical and operational owners maintain data pipelines, monitoring and incident response. The system should show its intended use and allow staff to report misleading alerts. A safe fallback is essential: if the model or data connection fails, staff continue standard clinical procedures.

Privacy and regulation

The presentation identifies consent and access controls through ABDM, the Digital Personal Data Protection Rules, CDSCO guidance for medical-device software and ICMR’s 2023 AI ethics guidance as relevant considerations. Precise legal obligations depend on the actual deployment and should be reviewed by qualified institutional teams. This report is an academic proposal, not legal advice.

Fairness and trust

Evaluate different patient groups and facility conditions. If performance is not reliable for a subgroup, report the limitation and consider mitigation or restricted use. Patients and health workers should understand that AI supports—not replaces—professional judgement. No cost figures are quoted because the presentation found no reliable public figures for the proposed system.

8. Appendix: Graphical Representation and Pilot Checklist

These figures summarise the group's proposed workflow, implementation plan and evaluation measures. They are conceptual graphics, not measured Apollo or national deployment results.

Appendix A — Proposed connected workflow

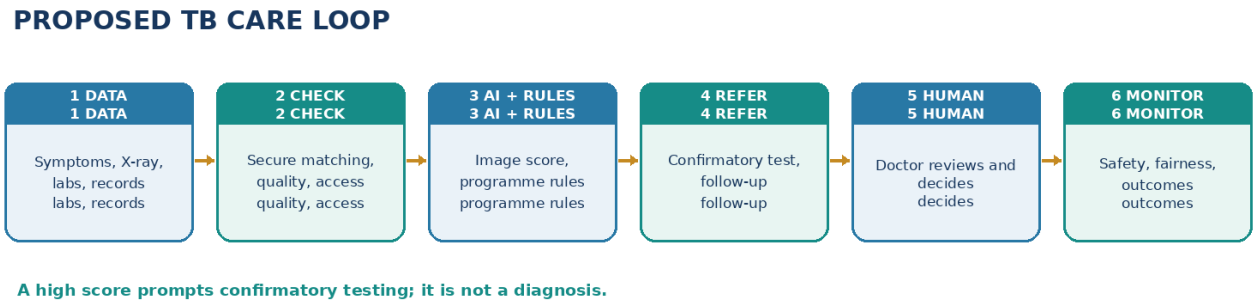


Figure 1. Data-to-decision loop. A high score prompts consideration of confirmatory testing; it does not establish a diagnosis.

Appendix B — Implementation roadmap

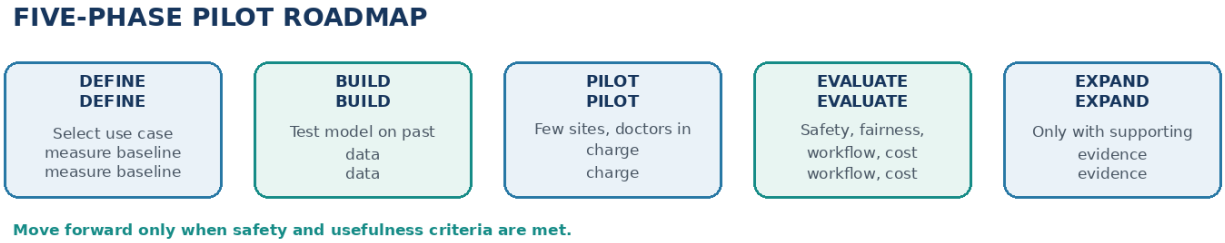


Figure 2. Five phases with decision gates before expansion.

Appendix C — Evaluation dashboard

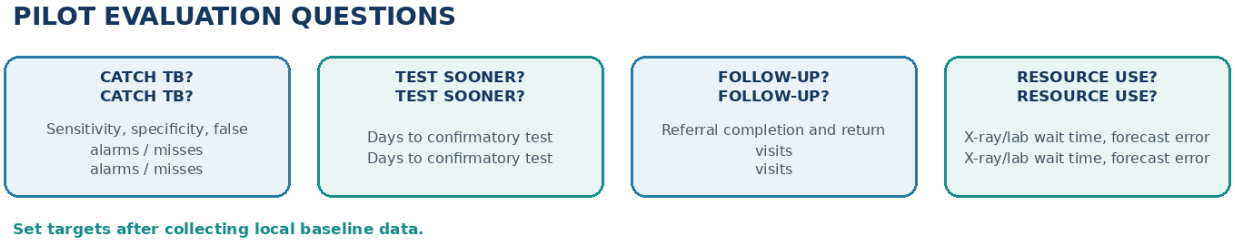


Figure 3. Evaluation questions; local baselines are needed before targets are set.

Appendix D — Pre-pilot checklist

- ■ Select districts and facilities with suitable digital X-ray and connectivity.
- ■ Define intended use, patient group, referral pathway and responsible clinician.
- ■ Document data sources, consent, access controls and downtime fallback.
- ■ Agree on model thresholds, confirmatory testing and escalation rules.
- ■ Measure baseline waiting times, referral completion and resource use.
- ■ Define subgroup checks, incident reporting and criteria to stop or expand.

References

Primary project source: Group 03 presentation, “AI in Healthcare — Apollo Hospitals, India,” supplied by the student, including speaker notes. Figures attributed to the presentation should be checked against original publications before formal submission.

World Health Organization. *Global Tuberculosis Report 2025.* Used in the presentation for India’s share of global TB cases and drug-resistant TB burden, and the change in incidence since 2015. Verify the exact table and wording in the original report.

Ministry of Health and Family Welfare, Government of India. *Rural Health Statistics 2021–22.* The presentation cites specialist-shortfall data for rural Community Health Centres, with data up to March 2022.

Ayushman Bharat Digital Mission (ABDM). Digital health records and ABHA-linked records information cited in the presentation as of July 2026. Verify the current official dashboard or publication for the exact figures before submission.

World Health Organization. Guidance on computer-aided detection for tuberculosis screening using chest radiography (2021). The presentation refers to WHO’s recommendation for computer-aided X-ray reading. Confirm exact title and edition from the WHO publication.

Indian Council of Medical Research (ICMR). (2023). *Ethical Guidelines for Application of Artificial Intelligence in Biomedical Research and Healthcare.*

Central Drugs Standard Control Organisation (CDSCO). Medical-device-software guidance referenced in the group presentation. Confirm the exact current guidance title and applicable requirements with official CDSCO sources.

Digital Personal Data Protection Act, 2023 and related Rules, as referenced in the presentation. Confirm applicable commencement and requirements from official Government of India publications.

Project examples in the supplied presentation: Apollo AICVD cardiac risk model; Siemens Healthineers AI-Rad Companion; Practo ABDM-compliant records software. These are examples of building blocks only, not endorsements or components of the group’s proposed integrated system.

Scope note: Sunita is fictional. The connected loop, unified programme rules, knowledge graph, follow-up tracking and cross-facility resource forecasting are proposals. No unsupported performance, cost or outcome figures have been added.